



Research Article

Electronic structure, magnetic properties and magnetocaloric effect of $\text{GdCo}_{2-x}\text{Ni}_x$ Gabriela Souca^{a,*}, Roxana Dudric^a, Ruxandra Cotop^a, Karsten Küpper^b, Coriolan Tiusan^{a,c}, Romulus Tetean^a^a Faculty of Physics, Babes-Bolyai University, Kogalniceanu 1, 400084 Cluj-Napoca, Romania^b University of Osnabrück, Fachbereich Physik, 49069 Osnabrück, Germany^c National Center of Scientific Research, France

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ABSTRACT

The electronic structure, magnetic properties, and magnetocaloric effect of $\text{GdCo}_{2-x}\text{Ni}_x$ with $x = 0.2, 0.25$, and 0.3 were investigated. X-ray diffraction measurements showed that all studied samples are single phase with the cubic MgCu_2 - Laves type structure. In agreement with experiments, the band structure calculations explain the following issues: the antiferromagnetic ground state of the system, the coupling between Gd and the transition metals (TM) mediated by the $5d$ electrons of Gd hybridized with the $3d$ electrons of the TM (Co, Ni), the reduction of the spin magnetic moment of Co when substituting with Ni due to Co-Ni hybridization. XPS measurements showed no significant shift in the binding energy of the investigated Co core levels with the change of Ni concentration. For all investigated samples the exchange splitting estimated to about 3.2 eV between the high spin final state and the low spin final state of the Co $3s$ core level gives direct evidence of the local magnetic moments on Co sites. The measured increase of the saturation magnetizations when substituting cobalt by nickel confirms the ferrimagnetic-type ordering. The decrease of the Co magnetic moments is related to Co-Ni hybridization and filling of bands with the extra-electrons provided by Ni with respect to Co. Moreover, the Arrott plots, the temperature dependence of Landau coefficients and the shapes of magnetic entropy changes confirm the presence of a second order magnetic phase transition. The $RCP(\Delta S)/\Delta B$ values decreases slowly when the Ni concentration increases and could be attributed to the narrowing of the $|\Delta S_m|$ peaks.

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1. Introduction

The RM_2 intermetallic compounds, where R is a rare-earth metal and M a $3d$ transition metal, have presented great interest for the researchers due to the fact that their crystal structure is simple enough to provide detailed investigations of basic phenomena, their interesting unique magnetic properties and wide technical applications [1,2]. These materials can be used as permanent magnets, magneto-optical storage media, for hydrogen storage and magnetostriuctive devices [1–5]. These types of compounds are also a good refrigerant because their large angular momentums of magnetic atoms lead to a large magnetic entropy change [6–8]. The magnetic properties of R-M_2 compounds can be subtly tuned by doping, which, through chemical pressure and/or band filling or depletion

effects can affect magnetic ordering, magnetic moments, and transition temperatures [6].

Intensely studied, GdCo_2 is a ferrimagnet with an opposite orientation of Gd and Co magnetic moments [9], exhibiting a second order magnetic phase transition [10] around the Curie point, T_C , which is located around 400 K [11]. Also, GdNi_2 is a ferrimagnet, having a low transition temperature, $T_C = 85$ K [12]. Until recently, the Ni magnetic moment in compounds with higher R concentration (RNi_2 , RNi , R_3Ni) was neglected. XAS measurement indicated that the $3d$ band of Ni in GdNi_2 is not completely occupied, and XMCD showed that it retains a finite magnetic moment of about $0.2 \mu_B$ and an angular momentum component [13].

The magnetic properties of $\text{Gd}(\text{Co}_{1-x}\text{Ni}_x)_2$ compounds were previously studied by Taylor [14], Burzo et al. [15] and T. Kaneko et al. [16]. Taylor has observed a strong decrease of the transition metal average magnetic moment for Ni concentrations higher than 0.5 . These results were explicated considering a rigid band model. According to Burzo et al. the transition metal magnetic moment

* Corresponding author.

E-mail address: gabrielasouca@gmail.com (G. Souca).

decreases monotonically when Ni concentration increases and this behavior was explained using the local environment model. On the other hand, Kaneko et.al. have report a similar behavior with Burzo of the transition metal moment, but their interpretation was based on the exchange-enhanced paramagnetism model [17]. Campbell model consider that the magnetic coupling between R and M atoms can be explained considering that the R-M exchange interactions are mediated by R5d band polarization [18]. Later on Burzo et.al have shown that the cobalt magnetic moment in RM₂ compounds is induced by 4f-5d-3d exchange path, the Co moment being stabilized by Co3d-Co3d short range exchange interactions with an additional induced polarization by R5d band [19].

From band structure calculation, it was shown, that the binding energies of the *d* bands near Fermi level are different, being larger in the case of GdNi₂ comparing with GdCo₂, showing that the hybridization interaction between Gd-*d* and Ni-*d* electrons is higher comparing to the case of Gd-*d* and Co-*d* electrons [20]. Electronic charge localization, in the down spin 3d orbital mostly, was shown in the transition metal calculated density of states. This localization implies the reduction of the total magnetic moment and the broadening of the 4 *f* level leaves [20].

In this paper we report a detailed correlated study of structural, electronic, and magnetic properties of GdCo_{1-x}Ni_x compounds. The dopant concentration was chosen to tune the transition temperature around 300 K. To get a deep insight on the physical properties of the transition metal, different analysis tools have been combined: magnetic measurements, X-ray photoelectron spectroscopy (XPS) and band structure calculations. In view of potential applications in magnetic refrigeration, a full evaluation of magnetocaloric properties around room temperature range was accomplished.

2. Experimental and computing methods

GdCo_{2-x}Ni_x samples were prepared with three different concentrations of Ni substituting atoms, *x* = 0.2, 0.25 and 0.3, by melting high purity (99.99 %) ingots in an arc furnace under purified argon atmosphere. Their homogeneity was ensured by several times consecutive melting. The samples were thermally treated in vacuum at 850 °C for 7 days and slowly cooled to room temperature. X-ray diffraction (XRD) was performed at room temperature on powder sample, using a Bruker D8 Advance AXS diffractometer with Cu K α radiation.

A 12 T VSM from Cryogenics was used for magnetic measurements, in the temperature range 5–600 K and external magnetic field up to 12 T. The magnetization isotherms registered between zero field and a maximum field (*H*₀) were used in order to calculate the magnetic entropy changes, using the thermodynamic relation:

$$\Delta S_m(T, H_0) = S_m(T, H_0) - S_m(T, 0) \\ = \frac{1}{\Delta T} \int_0^{H_0} [M(T + \Delta T, H) - M(T, H)] dH \quad (1)$$

with an increment in temperature between measured magnetization isotherms of $\Delta T = 5$ K. The refrigerant capacity or the relative cooling power (RCP), used to evaluate the magnetic refrigeration of materials, is calculated using the relation $RCP(S) = -\Delta S_m(T, H) \times \delta T_{FWHM}$, where ΔS_m is the maximum value of the magnetic entropy change and δT_{FWHM} its full-width at half-maximum.

The band structure calculations have been performed using the full potential linearized augmented plane-wave method (LAPW) +local orbitals (lo) as implemented in Wien2k ab-initio code [21] within a fully relativistic scheme. The Local Spin Density Approach +U Hubbard has been considered with an effective value of *U* = 0.58 Ry (7.88 eV) adjusted to match the calculated band structure with the position of the Gd-*f* level extracted from the photoemission

experiments (8.2 eV). This adjusted *U* value is in the range of values reported in the literature for GdTM₂ alloys [22,23]. The size of the basis set used in the wave function set expansion was $R_{MT} \cdot K_{max} = 7$, where *R*_{MT} is the smallest atomic muffin-tin sphere radius and *K*_{max} the largest *K*-vector, corresponding to a cutoff energy of 7.84 Ry. The integration in *k*-space used a modified tetrahedron method with 1000 *k*-points within irreducible Brillouin zone. A fully relativistic scheme has been considered, including spin-orbit interaction term in the Hamiltonian, for calculating spin and orbital magnetic moments of the atoms in the supercell model used to describe the GdCo_{1-x}Ni_x alloys with different concentrations.

The XPS spectra were recorded at room temperature using a PHI 5600ci ESCA spectrometer with monochromatized Al K α radiation. The pressure in the ultrahigh vacuum chamber was in the 10⁻¹⁰ mbar range during the measurements. The samples were cleaved in situ and the surface cleanness was checked by monitoring the O 1s and C 1s core levels in the survey spectra. Their intensity was very low, indicating a negligible contamination.

3. Results and discussions

3.1. Crystal structure

The XRD diffraction patterns of the investigated samples are presented in Fig. 1. The crystal structure and the lattice parameters were calculated by performing Rietveld refinements on the experimental data using the FullProf suite [24]. The theoretical peak positions and the differences between the experimental and calculated patterns are also shown in Fig. 1. In all cases only one phase was found, having the cubic MgCu₂ - Laves type structure. The Gd atoms occupies -43 m sites while Co and Ni are distributed randomly in -3 m positions. The calculated lattice parameters show a very small decrease, almost negligible with Co substitution by Ni, being 7.249(4) Å for *x* = 0.2, 7.247(8) Å for *x* = 0.25 and 7.244(7) Å for *x* = 0.3. This small deviation is somewhat expected, considering close values of atomic radii for cobalt (125 pm) and nickel (124 pm).

3.2. Band structure calculations

Using the FP-LAPW Wien2k computational code, to describe the GdCo_{1-x}Ni_x cubic Laves phase (C15) systems, a supercell model has

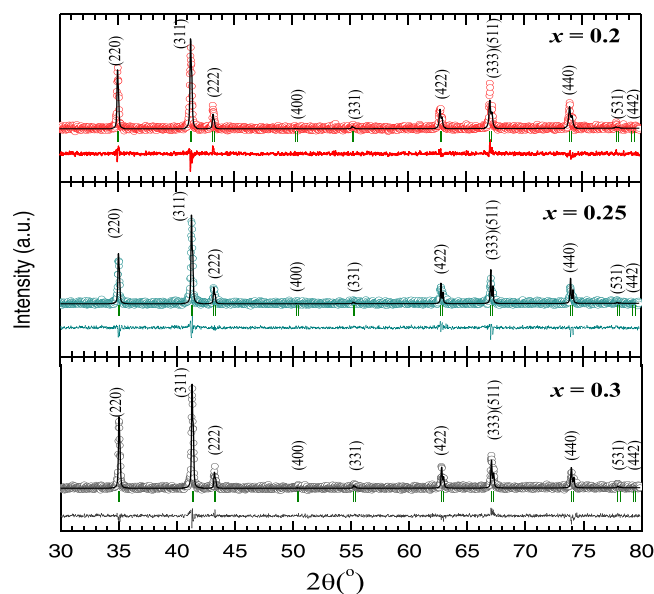


Fig. 1. X-ray diffraction patterns of the GdCo_{2-x}Ni_x compounds, where *x* = 0.2, 0.25 and 0.3.

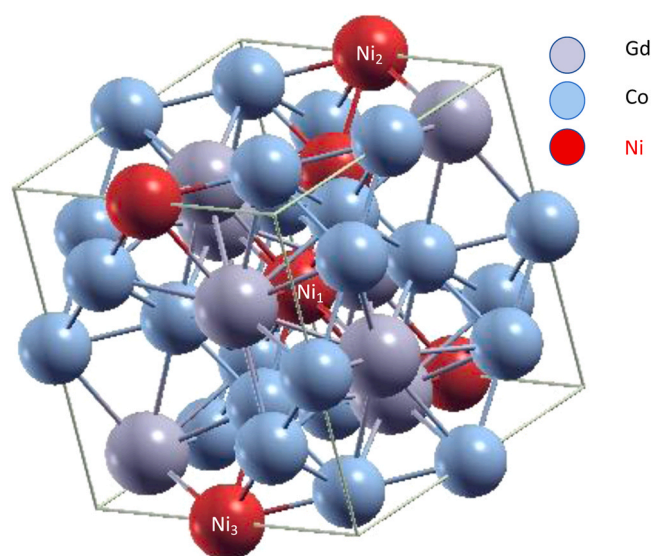


Fig. 2. The supercell model used in the band structure calculations of the $\text{GdCo}_{1.8}\text{Ni}_{0.2}$ compound, within the Wien2k FPLAPW LSDA+U framework. The substitutional Ni atoms can randomly occupy the Co positions. Here we illustrate the configuration where the Ni substitutionally occupy the $(1/2,1/2,1/2)$, $(1/2,3/4,3/4)$ and $(1/2,0,0)$ positions in the supercell.

been used. As seen in the model depicted in Fig. 2, the Gd atoms are arranged in a diamond structure consisted in two fcc lattices displaced by 0.25 of the diagonals and the transition metal atoms on site of the rhombohedral symmetry ($3m$) in a tetrahedral arrangement surrounded by four Gd next neighbors. The equivalent Gd atoms are located at $(1/8, 1/8, 1/8)$ $(7/8, 7/8, 7/8)$ and the transition metal Co (Ni) atoms occupy the $(1/2, 1/2, 1/2)$, $(1/2, 3/4, 3/4)$, $(1/2, 3/4, 3/4)$ $(3/4, 3/4, 1/2)$, $(3/4, 1/2, 3/4)$ positions. To describe the GdCo_2 compound, the supercell contains 24 atoms (8 Gd and 16 Co). Within the supercell, for different concentrations of Ni in the $\text{GdCo}_{1-x}\text{Ni}_x$ alloy, the Co is partially substituted by Ni. For instance, for the $\text{GdCo}_{1.8}\text{Ni}_{0.2}$ compound 3 Ni atoms substitute Co atoms in the supercell (Fig. 2). The lattice parameter of this last supercell is $a=b=c=7.2489 \text{ \AA}$, that minimizes the total energy of the supercell and the chosen values for the muffin-tin radius were: 2.5 a.u. for Gd and 2.29 a.u. for Ni and Co (a.u. = atomic Bohr units).

From the ab-initio calculations, we obtained the following main results:

(i) The ground-state of the system corresponds to an antiferromagnetic alignment of the spin magnetic moments of Gd and Co atoms. Total energy calculations show that the antiferromagnetic state is about 0.11 Ry lower than the ferromagnetic state.

(ii) The band structure and the total density of state for the $\text{GdCo}_{1.8}\text{Ni}_{0.2}$ compound are depicted in Fig. 3. One can see the position of the 4f localized Gd level, in both majority (occupied) and minority (unoccupied) spins. As mentioned, the effective U parameter has been adjusted to 0.58 Ry by matching the position of the occupied majority f-peak to the experimental XPS spectra where the f-peak was found to be located at about 8.2 eV below the Fermi level.

(iii) The calculated magnetic moments of the individual Gd, Co and Ni atoms are presented in the Table 1.

Considering the calculated total magnetic moment in the interstitial zone of $3.80 \mu_B$, the total magnetic moment per formula unit for the $\text{GdCo}_{1.8}\text{Ni}_{0.2}$ compound was found to be $5.52 \mu_B$. This value is slightly larger but in good agreement with the experimental value of $5.38 \mu_B$ (see Table 2). Furthermore, one can estimate an average magnetic moments of individual component atoms: $7.26 \mu_B$ for Gd, $-1.29 \mu_B$ for Co and $-0.29 \mu_B$ for the substitutional Ni. As reference, we also calculated the electronic properties of the GdCo_2 compound, for which the total magnetic moment per formula unit was found to

be smaller ($5.17 \mu_B$). Similar analysis leads to an average Gd spin moment of $7.27 \mu_B$ and Co spin moment of $-1.31 \mu_B$ Co, respectively.

Similar calculations have been performed on other possible supercell substitutional configurations. Differences in the individual atom projected moment related to different local environment of atom in the supercell are averaged to similar average magnetic moments for Gd, Co, Ni and total magnetic moment per formula unit. However, for a better statistical description of the shape of the density of states in a disordered alloy phase, random atomic positions configurations can be generated using statistical tools (e.g. Special Quasirandom Structures formalism). Averaging the calculated DOS “smoots” local peaks related to the ordered alloy model we involved. The DOS shape will then better match to the one issued from KKR-CPA methods for disordered alloys.

The band structure analysis allows us to get deeper in the mechanisms responsible on the magnetic moments of each element in the compounds. First, a larger calculated Gd moments with respect to the atomic expectation ($7 \mu_B$) clearly comes from the hybridization of the Gd-5d and Co-3d orbitals: from the spin resolved valence charge distribution analysis we extract exactly the additional $0.26 \mu_B$ contribution of the hybrid 5d-Gd orbital to the total Gd spin magnetic moment. The coupling between Gd and the transition metals (TM) is therefore mediated by the 5d electrons of Gd hybridized with the 3d electrons of the TM (Co, Ni). Second, the $\text{GdCo}_{1.8}\text{Ni}_{0.2}$ density of states analysis reveals the main mechanisms responsible of the reduction of the spin magnetic moment of Co when doping with Ni. They are mainly related to Co-Ni hybridization and not simply to a rigid band energy downward displacement effect, due to the extra-electron brought by the substitutional doping Ni in the Co host places.

The orbital projected DOS analysis (Fig. 4) clearly illustrates that the minority 3d-bands of Co and Ni are strongly hybridized, the shape of each element projected DOS being similar in all the energy range. For the majority spin, the hybridization is less strong, the Co and Ni atoms bring their individual contributions to the total density of states and Ni brings an additional electron to the band filling with respect to Co and the rigid band concepts remains more suitable.

On the other hand, further theoretical band-structure calculations, considering the spin-orbit interactions contribution in the relativistic Hamiltonian, allows to estimate the orbital moments corresponding to the magnetization oriented along the z axis. For the Gd, a negligible orbital moment of $-0.003 \mu_B$ (antiparallel with the spin moment) was found. This agrees with the Hund rules; for its 7-f electrons in the atomic configuration the maximum spin angular moment is attained for $m_l = 0$, corresponding to a zero projection of the orbital magnetic moment along z. Similarly, small, or negligible orbital magnetic moments were also found for both Co ($-0.11 \mu_B$) and Ni ($-0.03 \mu_B$), both parallel with the corresponding spin moment.

3.3. XPS

XPS core level spectra provide important information on the electronic structure, especially in the case of 3d transition metal ions. For systems with unpaired electrons in the valence levels, the interaction of the core hole with the valence electrons leads to a multiplet (exchange) splitting of the core level lines. In the simplest model this splitting originates from the intra-atomic exchange coupling between the photo-hole and the unpaired 3d electrons in the final state of the system and hence the magnitude of the splitting ΔE_{ex} will reflect the net spin in the valence band. However, we have to take into account the effect of a core-hole screening in the final state of the photoemission and the charge-transfer effects between the 3d metal ion and the ligand atom. The multiplet splitting of the 3s line for 3d transition metals with a local magnetic moment is a good measure of the local moment of the ground state only when the charge-transfer satellite in the 2p core level spectra is small.

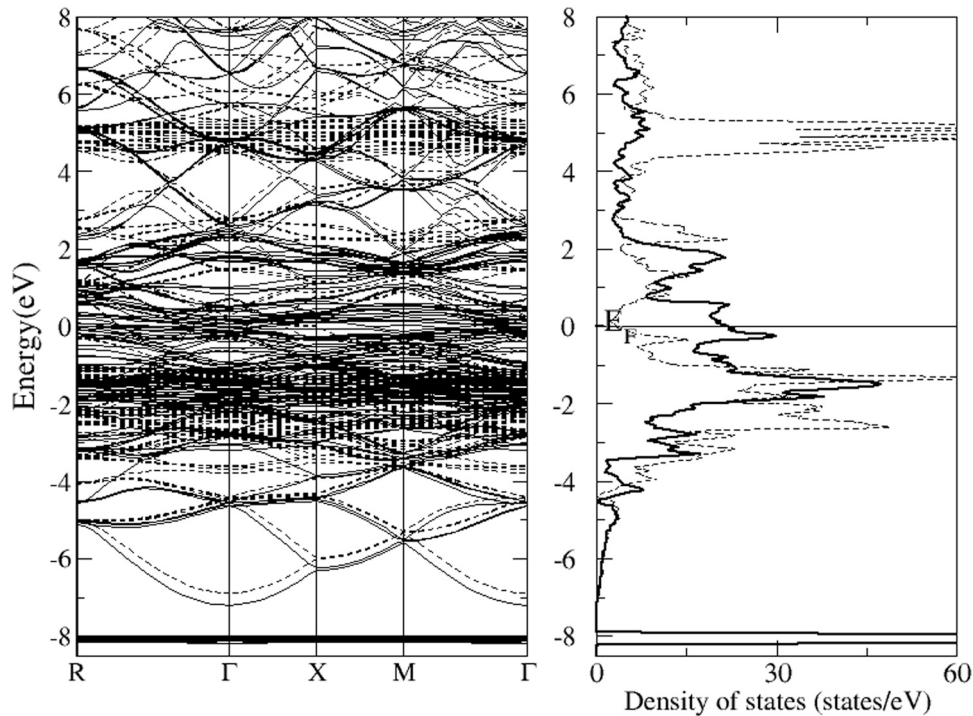


Fig. 3. Band structure (left) and corresponding total density of states for the $\text{GdCo}_{1.8}\text{Ni}_{0.2}$ compound. The localized Gd-f states are located at -8.2 eV below E_F (majority spin) and $+5$ eV above E_F (minority spin).

Table 1
Calculated atom projected magnetic moments and position in the supercell.

Atom	Position in supercell	Spin magnetic moment (μ_B)
Gd	(1/8, 1/8, 1/8) (7/8, 7/8, 7/8)	7.26
	(1/8, 5/8, 5/8) (7/8, 3/8, 3/8)	
	(5/8, 5/8, 1/8) (3/8, 3/8, 7/8)	7.25
	(5/8, 1/8, 5/8), (3/8, 7/8, 3/8)	
Ni	(1/2, 1/2, 1/2) (1/2, 3/4, 3/4) (1/2, 0, 0)	0.29
Co	(3/4, 3/4, 1/2) (3/4, 1/2, 3/4)	-1.26
	(1/4, 0, 3/4) (1/4, 3/4, 0)	
	(0, 0, 1/2) (0, 1/2, 0)	-1.28
	(1/4, 1/4, 1/2) (1/4, 1/2, 1/4)	-1.31
	(3/4, 1/4, 0) (3/4, 0, 1/2)	
	(0, 1/4, 3/4) (0, 3/4, 1/4)	-1.33

Table 2
The values of the Curie temperature, saturation magnetization, and magnetic moments in $\text{GdCo}_{2-x}\text{Ni}_x$ compounds. The values of $7.26 \mu_B$ for Gd moment and $-0.29 \mu_B$ for Ni moment were extracted from ab-initio calculations -see Section 3.2.

x	T_C (K)	M_0 ($\mu_B/\text{f.u.}$)	m_{Gd} (μ_B/atom)	m_{Ni} (μ_B/atom)	m_{Co} (μ_B/atom)
0.2	332	5.38	7.26	-0.29	-1.01
0.25	313	5.50	7.26	-0.29	-0.96
0.3	303	5.68	7.26	-0.29	-0.88

Fig. 5 shows the Co 2*p* and Co 3*s* core-level spectra of the investigated compounds. There is no significant shift in the binding energy of the investigated Co core levels with the change of Ni concentration, as expected considering that the lattice parameter and hence the distance between the atoms is almost the same for the investigated $\text{GdCo}_{2-x}\text{Ni}_x$ compounds. The Co 2*p* XPS spectra are dominated by two features located at 778.2 eV and 793.2 eV attributed to the 2*p*_{3/2} and 2*p*_{1/2} states. Both these states are accompanied by broad satellite lines at higher binding energies, which may be explained by assuming more than one 3*d*^{*n*} final state configuration.

The Co 3*s* core level spectra show a main line located at about 101 eV and a satellite at higher binding energy, due to the exchange interaction J_{dc} between the core hole spin *s* and the 3*d* electron spin *S* in all investigated compounds [25]. In order to evaluate the exchange splitting, the XPS spectra were fitted in the 90–120 eV binding energy region using three components: two Co 3*s* peaks, corresponding to the high spin final state *S* + 1/2 and the low spin final state *S* - 1/2, respectively, and one for the Ni 3*s* core level, which is situated at about 110.7 eV. The curve fitting results are presented in Fig. 6. For all investigated samples we found an exchange splitting of about 3.2 eV between the high spin final state and the low spin final state of the Co 3*s* core level, giving a direct evidence of the local magnetic moments on Co sites. This value is smaller to that found in the pure Co metal, namely 4.5 eV [26]. Since the exchange splitting $\Delta E_{\text{ex}} = J_{dc}(2S + 1)$ is proportional with the Co local moment [27], we estimate an average magnetic moment for Co ions of about 1.2 μ_B in all investigated $\text{GdCo}_{2-x}\text{Ni}_x$ compounds.

3.4. Magnetic measurements

The magnetization measurements as a function of temperature, in the applied magnetic field of 0.05 T are presented in Fig. 7. All the prepared compounds undergo a transition from the paramagnetic state to the ordered magnetic state. The Curie temperatures were calculated as the minima in the first derivative of magnetization dM/dT vs temperature and plotted in the inset of Fig. 7. No other magnetic transitions are present in the investigated cases, confirming the high quality of the samples. As the concentration of Ni increases, the Curie temperature, T_C , significantly gets reduced towards room temperature as can be seen in Table 2. Three types of interaction have to be considered *R*-*R*, *R*-*M* and *M*-*M*. The *R*-*R* interactions are of RKKY-type being indirect, weak and occurring between the 4*f* electrons of the rare-earth atoms [4]. The *M*-*M* interactions are direct, being the strongest ones, and take place between the 3*d* shells of the *M* atoms [4]. The strength of *R*-*M* exchange interactions are intermediate between *R*-*R* and *M*-*M* ones, and appear between the

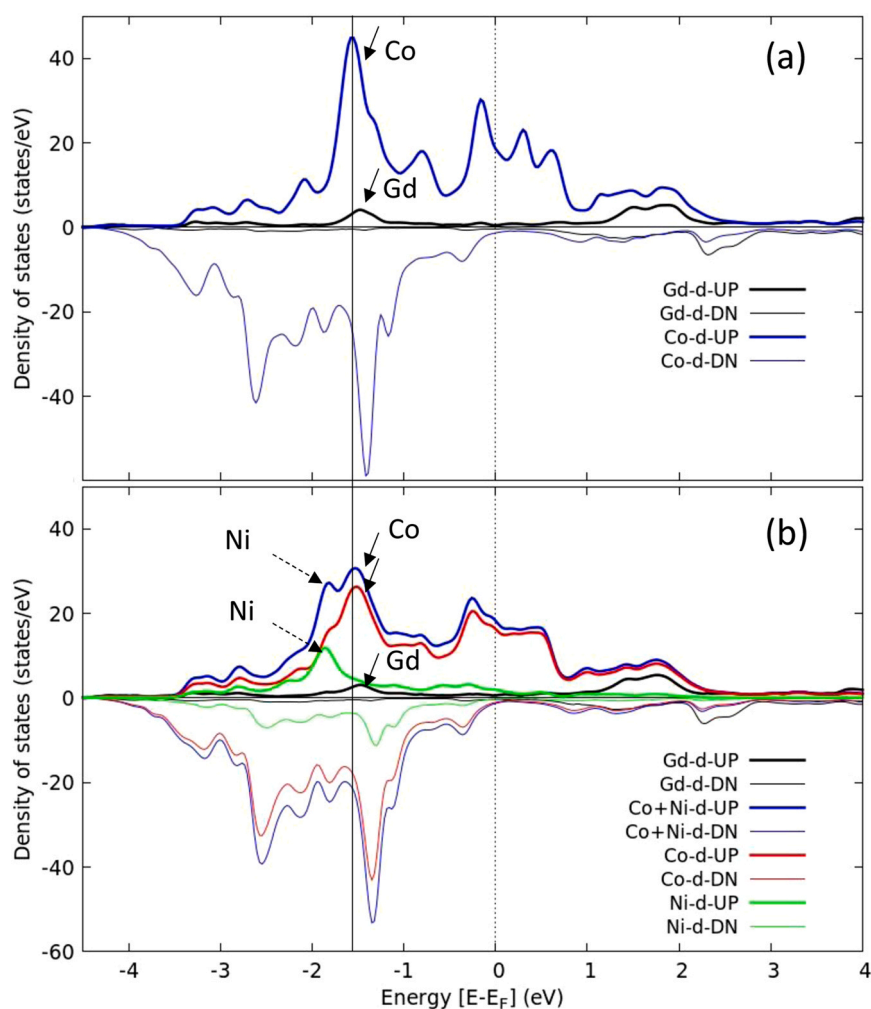


Fig. 4. Orbital projected (d) density of states for the (a) reference GdCo_2 and (b) $\text{GdCo}_{1.8}\text{Ni}_{0.2}$ compounds. The contribution of each element is indicated with different color (see legend). Continuous line arrows indicate for both compounds the Gd-Co hybridization of d-states (e.g. peaks in Gd and Co d-DOS at about -1.6 eV below E_F) and in (b) the Ni (dashed arrow) and Co (continuous line arrow) contribution to the majority DOS.

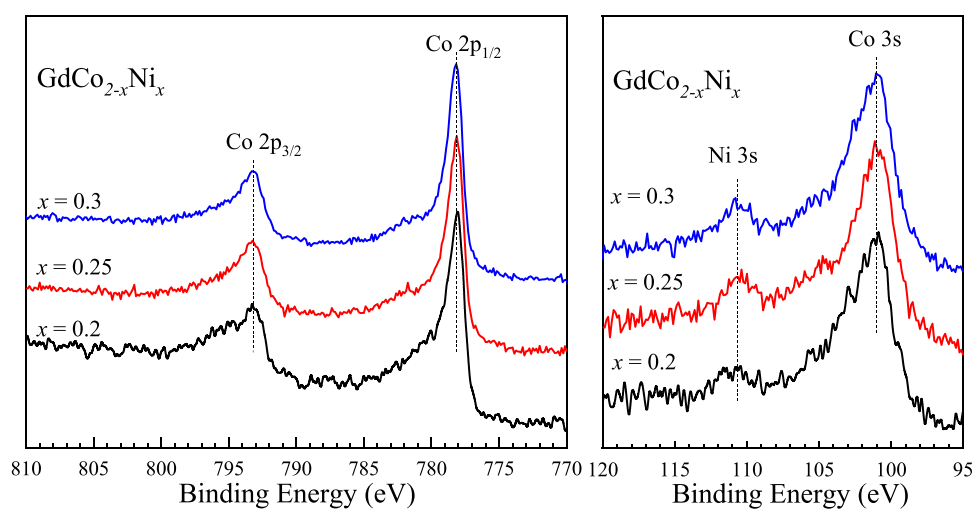


Fig. 5. Co 2p (right) and Co 3s (left) XPS spectra of $\text{GdCo}_{2-x}\text{Ni}_x$ compounds, where $x = 0.2, 0.25$ and 0.3 .

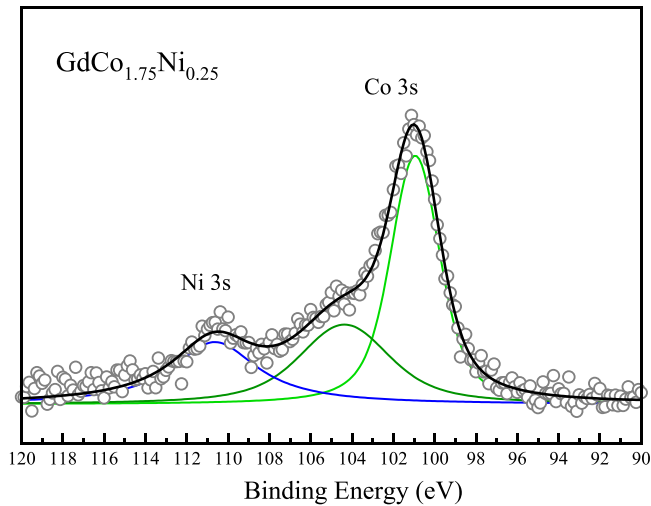


Fig. 6. Curve fitting results for the XPS spectrum of GdCo_{1.75}Ni_{0.25} (open circles correspond to the experimental spectrum and the continuous curves to the fitting results, after background subtraction).

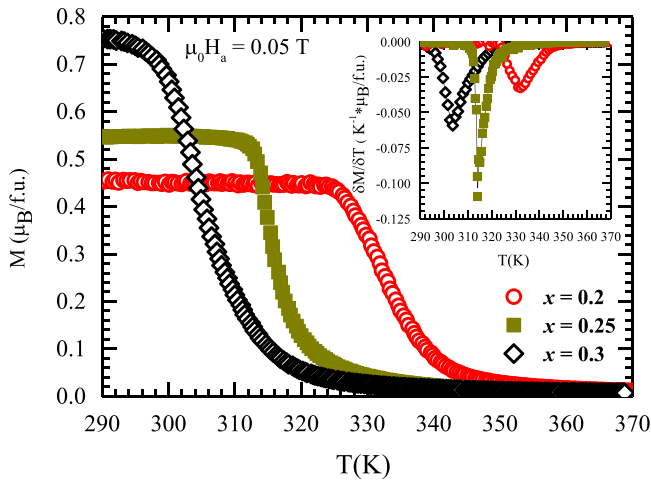


Fig. 7. The magnetization vs temperature for GdCo_{2-x}Ni_x compounds, in the applied magnetic field of 0.05 T. The inset shows the first derivative of magnetization vs temperature.

polarized R 5d shells M 3d shells [28]. The Curie temperature is determined, mainly, by the M - M strength interaction which depends both on the interatomic distances and the number of nearest neighbors [28].

In our case the substitution of Co by Ni modifies the exchange Co3d - Co3d interactions (Co3d - Ni3d or Ni3d - Ni3d being weaker), enabling the decrease of T_C toward room temperature. Therefore, the almost inversely proportional decrease of Curie temperature on Ni concentration value can be attributed to the magnetic dilution.

The magnetization isotherms recorded at 4.2 K are plotted in Fig. 8. Values of $4.96 \mu_B/f.u.$ [15], $5 \mu_B/f.u.$ [29] $4.89 \mu_B/f.u.$ [30] were reported previously for the saturation magnetization in the parent compound, GdCo₂. The saturation magnetization values obtained for GdCo_{2-x}Ni_x compounds are higher than those reported, being between $5.38 \mu_B/f.u.$ for $x = 0.2$ and $5.68 \mu_B/f.u.$ for $x = 0.3$ (Table 2). The fact that the saturation magnetization increases when replacing cobalt by nickel is in good agreement with a ferrimagnetic-type

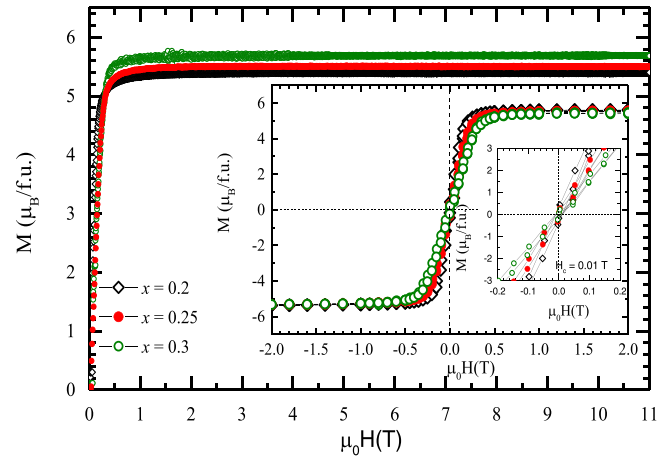


Fig. 8. Magnetic isotherms of GdCo_{2-x}Ni_x compounds at 4.2 K in the applied magnetic field up to 10 T. The inset shows the magnetic hysteresis for the investigated samples at 4.2 K.

ordering, gadolinium and cobalt magnetic moments being anti-parallelly oriented.

Previously it was reported that nickel is not magnetic in RNi₂ compounds [31]. A magnetic moment of $0.2 \mu_B/Ni$ atom was reported on GdNi₂ compound by Mizukami et al. from magnetic circular dichroism studies [13]. The calculated magnetic moments for Gd and Ni seem not to depend on Ni concentration. In order to extract the Co magnetic moments from experimental data we have considered the calculated magnetic moments for Gd and Ni atoms, as listed in Table 1. The values of the Curie temperature, saturation magnetization, and magnetic moments are shown in Table 2.

The substitution of Co by Ni modifies the strength of 3d-3d exchange interactions, being smaller in Co 3d-Ni 3d than in Co 3d-Co 3d. This would roughly explain the decrease of the Curie temperature when substituting the Co with Ni. Moreover, the Ni substitution slightly lowers the Co magnetic moment in GdCo_{2-x}Ni_x compounds. The mechanisms of the cobalt magnetic moment decrease are deeper revealed by band structure analysis (see Section 3.2). It is mainly related to Co-Ni hybridization and filling of bands with extra-electrons provided by Ni that has one more electron with respect to Co.

On the other hand, the analysis of magnetization curve shows a very small coercive field at 4 K, for all concentration of Ni in the GdCo_{2-x}Ni_x compounds, (see inset of Fig. 8). This indicates that, in these compounds, the magnetocrystalline anisotropy is negligible. This experimental result perfectly agrees with the previous theoretical calculations results of Gavrilenco and Wu [32] which illustrated that the magnetocrystalline energy linearly varies with the lattice distortion of the unit cell (compressive or tensile stress), changing the sign and crossing the zero around the equilibrium theoretical lattice parameter of the GdCo₂ compounds that corresponds in our case to $a_{lat} = 13.6988$ Bohr.

The analysis of Arrott plots depicted in Fig. 9 highlights the presence of a second order magnetic phase transition for the compounds with $x = 0.2$ and $x = 0.3$, respectively. Similar behavior was obtained for the sample with $x = 0.25$.

The Arrott plots were calculated from the magnetic isotherms around transition temperature in the temperature range 175–380 K. No inflexion point at transition temperature was found for all GdCo_{2-x}Ni_x compounds, which is a behavior typical for a second-order magnetic type transition. In order to confirm the transition type we

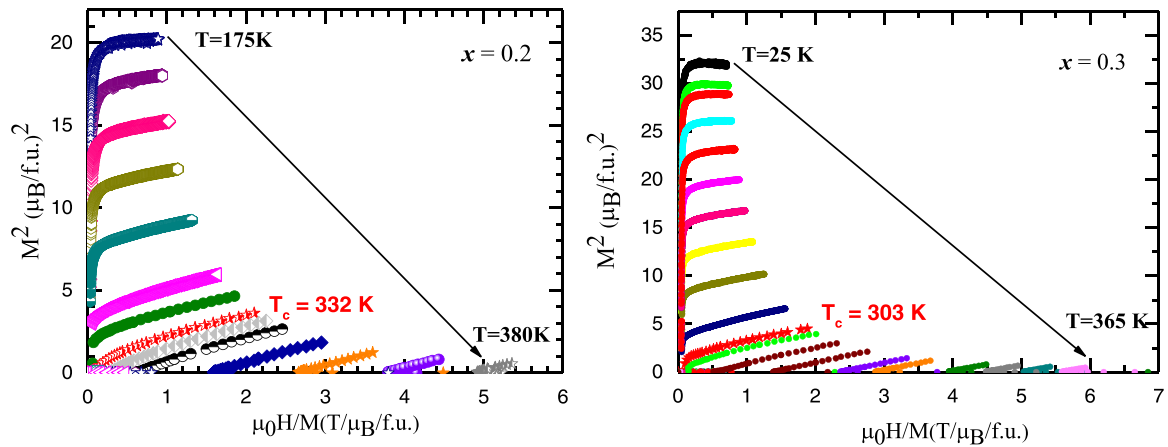


Fig. 9. Arrott plots for the compounds with $x = 0.2$ and $x = 0.3$.

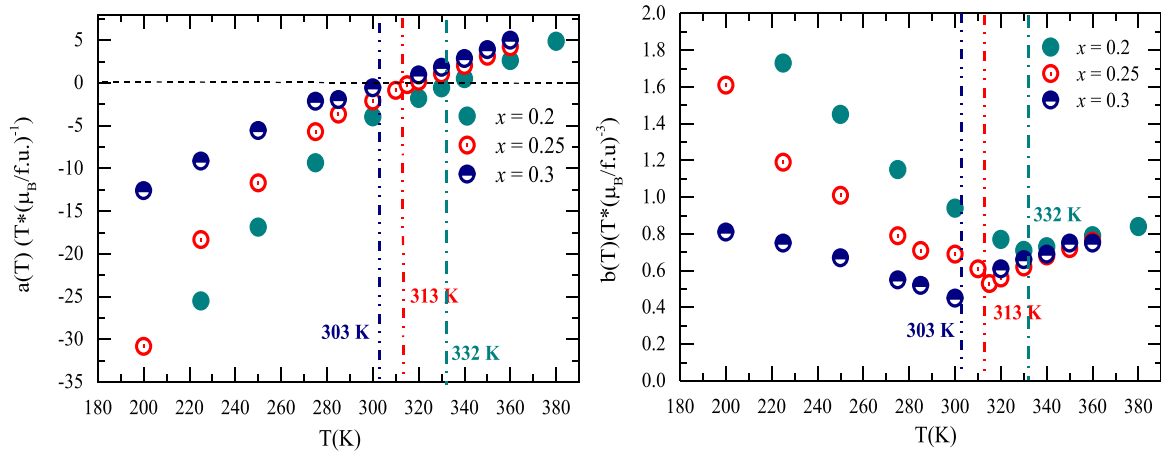


Fig. 10. The temperature dependences of the Landau a and b coefficients for the studied samples.

have consider the Landau expression for the magnetic free energy [33]:

$$F = \frac{1}{2}a(T)M^2 + \frac{1}{4}b(T)M^4 + \frac{1}{6}c(T)M^6 - \mu_0HM \quad (2)$$

where $a(T)$, $b(T)$ and $c(T)$ are so called Landau coefficients and their temperature dependences shows the type of the magnetic phase

transition. Landau coefficients can be determined from the relation between M and H :

$$\mu_0H = a(T)M + b(T)M^3 + c(T)M^5 \quad (3)$$

The magnetic transition order is given by the $b(T)$ sign, being of first order if $b(T)$ at the transition temperature is negative and of second order if the $b(T) \geq 0$. The temperature dependences of the Landau coefficients were calculated by fitting each magnetization isotherms with the Eq. (3) and are presented in Fig. 10. One can see that in all cases the $a(T)$ coefficient is changing the sign at the transition temperature, as was predicted by Landau theory, while the $b(T)$ coefficient is positive and have the minima at $T = T_c$. The values obtained for $b(T)$ coefficient confirm the observations from the Arrott plots and prove that the phase transition is of second order.

Magnetic hysteresis loops recorded at room temperature (Fig. 11), in the applied field of -2 T and 2 T show a negligible hysteresis for the samples with $x = 0.2$ and $x = 0.25$ and the lack of hysteresis curve for the sample with $x = 0.3$, which is favorable for magnetocaloric materials. In order to have a reversible MCE, no magnetic hysteresis loss is required. Therefore, the investigated samples are a promising candidate for magnetic refrigeration applications.

The magnetic entropy change was calculated based on the magnetization isotherms measured between 280 K and 380 K, with a step of 5 K, in the external magnetic field between 0 and 2 T and respectively $0-4$ T and plotted as a function of temperature in Fig. 12. The maximum value of the magnetic entropy change is reached at temperatures close to the magnetic transition ones, for all the investigated samples. The value of the maximum entropy change

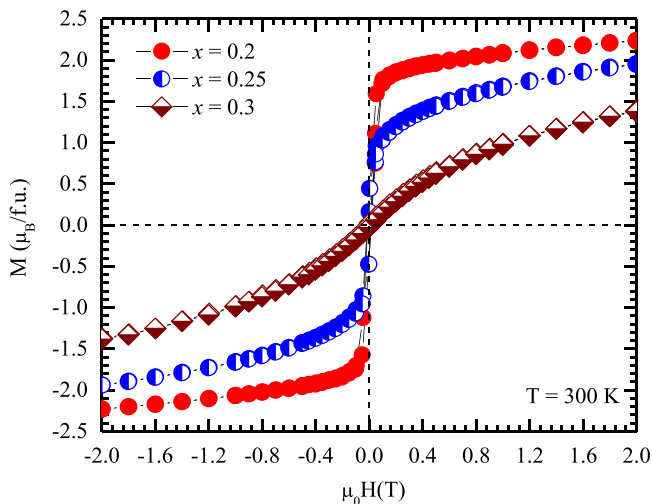


Fig. 11. Magnetic hysteresis loops recorded at $T = 300$ K for $\text{GdCo}_{2-x}\text{Ni}_x$.

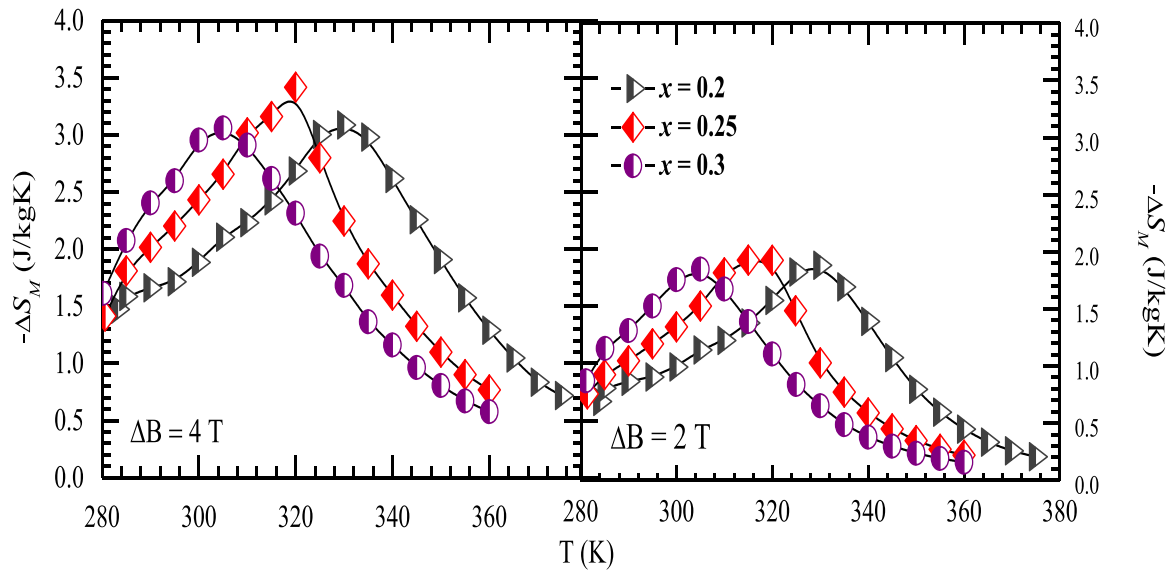


Fig. 12. The temperature dependences of the magnetic entropy change for $\text{GdCo}_{2-x}\text{Ni}_x$.

depends on the sharpness of the magnetic transition. In our case the sharper transition is for the sample with $x=0.25$ and one can see that for this sample the magnetic entropy change is maximum. On the other hand, the Ni-doping decreases the exchange interactions and has a more important contribution on the values of the temperatures where ΔS has maxima. The substitution of Co by Ni causes a very small increase in maximum entropy change values, as listed in Table 3. Nevertheless, the $|\Delta S_m|$ peaks are broader, which is mandatory for a good refrigerant material. Moreover, it can be observed that the curves are nearly symmetric distributed around the transition temperature, a behavior generally characteristic for materials exhibiting a second order magnetic phase transition [34].

Similar MCE were reported previously on $\text{Ho}_{1-x}\text{Gd}_x\text{Co}_2$ compounds, with $|\Delta S_m|$ values of 3.42, 3 and 3.08 J/(kg K) for $x=0.5$, 0.6 and 0.8 in the external magnetic field of 5 T [35].

The magnetic refrigeration efficiency was evaluated by calculating the relative cooling power (RCP) for all the investigated samples, larger values corresponding to better magnetocaloric materials [6]. The $\text{RCP}(\Delta S)$ values, presented in Table 3, are rather large, indicating that the materials could be used for the magnetic refrigeration applications.

It was reported that the magnetic field dependence of the magnetic entropy change for the compounds which present a second order type transition follows a master curve, which can be expressed as $\Delta S_M \propto B^n$, where n has a value of 0.75 for $T=T_C$ and a value of 2 in the paramagnetic region [36]. If the magnetic applied field is changing from B_1 to $B_2 = 2B_1$, the maximum magnetic entropy change value corresponding to a field change from 0 to B_2 will be $\Delta S_M(B_1) = 0.59\Delta S_M(B_2)$. In $\text{GdCo}_{2-x}\text{Ni}_x$ compounds the values of the factor $\Delta S_M(B_2)/\Delta S_M(B_1)$ are 0.59 for $x=0.2$, 0.58 for $x=0.25$ and 0.60 for $x=0.3$

when the applied magnetic field change is reduced from 0 to 4 T to 0–2 T, being in very good agreement with the behavior of the reported master curve [36].

For technical application it is better to use another characteristic parameter, related to the external field variation, namely the specific renormalized cooling power, $\text{RCP}(\Delta S)/\Delta B$ (Table 3) [37]. For our samples $\text{RCP}(\Delta S)/\Delta B$ values decrease slowly when the Ni concentration increases but do not depend significantly on the applied magnetic field change. The decrease of the $\text{RCP}(\Delta S)/\Delta B$ values with increasing Ni concentration could be attributed to the narrowing of the $|\Delta S_m|$ peaks.

In order to check the suitability, efficiency and utility of these compounds for magnetocaloric cooling we have calculated the Temperature-averaged Entropy Change, TEC [38]. It was shown that TEC provides an indicator to choose the most promising material for a specific application [38]. The TEC values are calculated over a range of temperatures, ΔT_{H-C} (the difference between the hot and cold heat exchangers required for a device). TEC values can be calculated using the $\Delta S_M(T)$ values:

$$\text{TEC}(\Delta T_{H-C}) = \frac{1}{\Delta T_{H-C}} \max \left\{ \int_{T_{\text{mid}} - \frac{\Delta T_{H-C}}{2}}^{T_{\text{mid}} + \frac{\Delta T_{H-C}}{2}} |\Delta S_M(T)| dT \right\} \quad (4)$$

where T_{mid} is the temperature that maximizes the $\text{TEC}(\Delta T_{H-C})$ for a given ΔT_{H-C} . In our investigations calculations were made for $\Delta B = 2$ T or 4 T and ΔT_{H-C} are fixed between 5 and 50 K with a step of 5 K for all studied samples. The ΔT_{H-C} dependences of TEC are shown in Fig. 13. The highest value was obtained for the sample with $x=0.25$ for which the magnetic entropy change is higher. The calculated

Table 3
The magnetocaloric parameters for $\text{GdCo}_{2-x}\text{Ni}_x$ compounds.

x	T_{max} (K)	$ \Delta S_m $ (J/kgK)		δT_{FWHM} (K)		$\text{RCP}(\Delta S)$ (J/kg)		$\text{RCP}(\Delta S)/\Delta B$ (J/kgT)	
		$\Delta B=4$ T	$\Delta B=2$ T	$\Delta B=4$ T	$\Delta B=2$ T	$\Delta B=4$ T	$\Delta B=2$ T	$\Delta B=4$ T	$\Delta B=2$ T
0.2	330	3.07	1.82	71	55	217.97	100.10	54.49	50.05
0.25	319	3.43	2.00	59	46	202.37	92.00	50.59	46.00
0.3	304	3.07	1.84	46	44	141.22	80.96	35.30	40.48

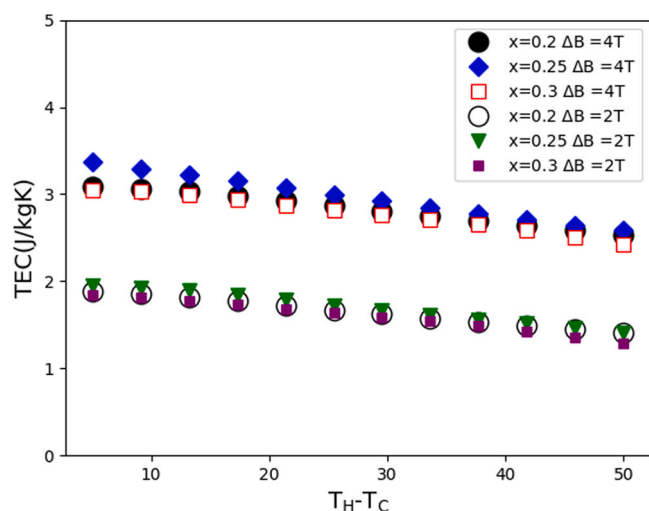


Fig. 13. The temperature-averaged Entropy Change, TEC coefficient, calculated for 3 concentrations: $x=0.2$, 0.25 , 0.3 and $\Delta B=2$ T or 4 T (see legend).

values are around 3.3 J/Kg K and around 2 J/kgK for $\Delta B=4$ T respectively 2 T, very close with other compounds showing the potential of these material for applications. One can see that the TEC values decrease with increasing T_H-T_C range for all sample in the same way as was reported earlier [39–41].

4. Conclusions

A detailed analysis of electronic and magnetic properties of $\text{GdCo}_{2-x}\text{Ni}_x$ compounds, with $x=0.2$, 0.25 and 0.3 has been performed, involving different tools: structural analysis, electronic structure calculations combined with XPS, magnetic and magnetocaloric characterization. X-ray diffraction measurements showed that all studied samples are single phase with the cubic MgCu_2 - Laves type structure. The Gd occupies ~ 43 m sites while Co and Ni are distributed randomly in ~ 3 m positions. The lattice parameters show a very small decrease, almost negligible with substitution of Co by Ni, somewhat expected, considering close values of atomic radii for cobalt and nickel.

Band structure calculations reveal the following main features: (i) The ground-state is antiferromagnetic, with 0.11 Ry lower energy than the ferromagnetic state (ii) The total magnetic moment per formula unit for the $\text{GdCo}_{1.8}\text{Ni}_{0.2}$ compound of $5.52 \mu_B$, in good agreement with the experimental value of $5.38 \mu_B$ and the atomic average magnetic moments are: Gd ($7.26 \mu_B$), Co ($-1.29 \mu_B$) and Ni ($-0.29 \mu_B$). The larger calculated Gd moments with respect to the atomic expectation ($7 \mu_B$) was explained by the hybridization of the Gd- $5d$ and Co- $3d$ orbitals. (iii) The reduction of the spin magnetic moment of Co when doping with Ni is related to Co-Ni hybridization and not simply to a rigid band energy downward displacement effect, due to the extra-electron brought by the substitutional doping Ni in the Co host places.

XPS measurements showed no significant shift in the binding energy of the investigated Co core levels with the change of Ni concentration, as expected considering that the lattice parameter and hence the distance between the atoms is almost the same for investigated $\text{GdCo}_{2-x}\text{Ni}_x$ compounds. The Co $2p$ XPS spectra are dominated by two features located at 778.2 eV and 793.2 eV attributed to the $2p_{3/2}$ and $2p_{1/2}$ states. Both these states are accompanied by broad satellite lines at higher binding energies, which may be explained by assuming more than one $3d^n$ final state configuration. The Co $3s$ core level spectra show a main line located at about 101 eV and a satellite at higher binding energy, due to the exchange interaction between the core hole spin s and the $3d$ electron spin S in

all investigated compounds. For all investigated samples was found an exchange splitting of about 3.2 eV between the high spin final state and the low spin final state of the Co $3s$ core level, giving direct evidence of the local magnetic moments on Co sites. An average magnetic moment about $1.2 \mu_B/\text{Co atom}$ was estimated.

Magnetic measurements reveal a decrease of the Curie temperature when Ni concentration increase, explained by changes of the exchange Co $3d$ - Co $3d$ interactions. The saturation magnetization increases when replacing cobalt by nickel, as expected for a ferrimagnetic-type ordering. Correlated with band-structure analysis, the measured decrease of Co magnetic moments has been related to Co-Ni hybridization and filling of bands with extra-electrons provided by Ni.

Arrott plots highlight the presence of a second order magnetic phase transition. In all cases the $a(T)$ Landau coefficient is changing the sign at the transition temperature, as was predicted by Landau theory, while the $b(T)$ Landau coefficient is positive confirming the observations from the Arrott plots and prove that the phase transition is of second order. The substitution of Co by Ni causes a very small increase in maximum entropy change values. The $|\Delta S_m|$ peaks are broader, and the curves are nearly symmetric distributed around the transition temperature, a behavior generally characteristic for a second order magnetic phase transition. The $RCP(\Delta S)/\Delta B$ values decreases slowly when the Ni concentration increases but do not depend significantly on the applied magnetic field change. The decrease of the $RCP(\Delta S)/\Delta B$ values with increasing Ni concentration could be attributed to the narrowing of the $|\Delta S_m|$ peaks. The TEC values are high enough in order to consider these compounds for magnetocaloric cooling applications.

CRediT authorship contribution statement

Gabriela Souza: Data curation, Investigation, Writing – original draft, Writing – review & editing. **Roxana Dudric:** Data curation, Investigation, Writing- Reviewing. **Ruxandra Cotop:** Data curation, Investigation. **Karsten Küpper:** Data curation, Investigation. **Coriolan Tiusan:** Data curation, Investigation, Methodology, Software, Writing- Reviewing. **Romulus Teteian:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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